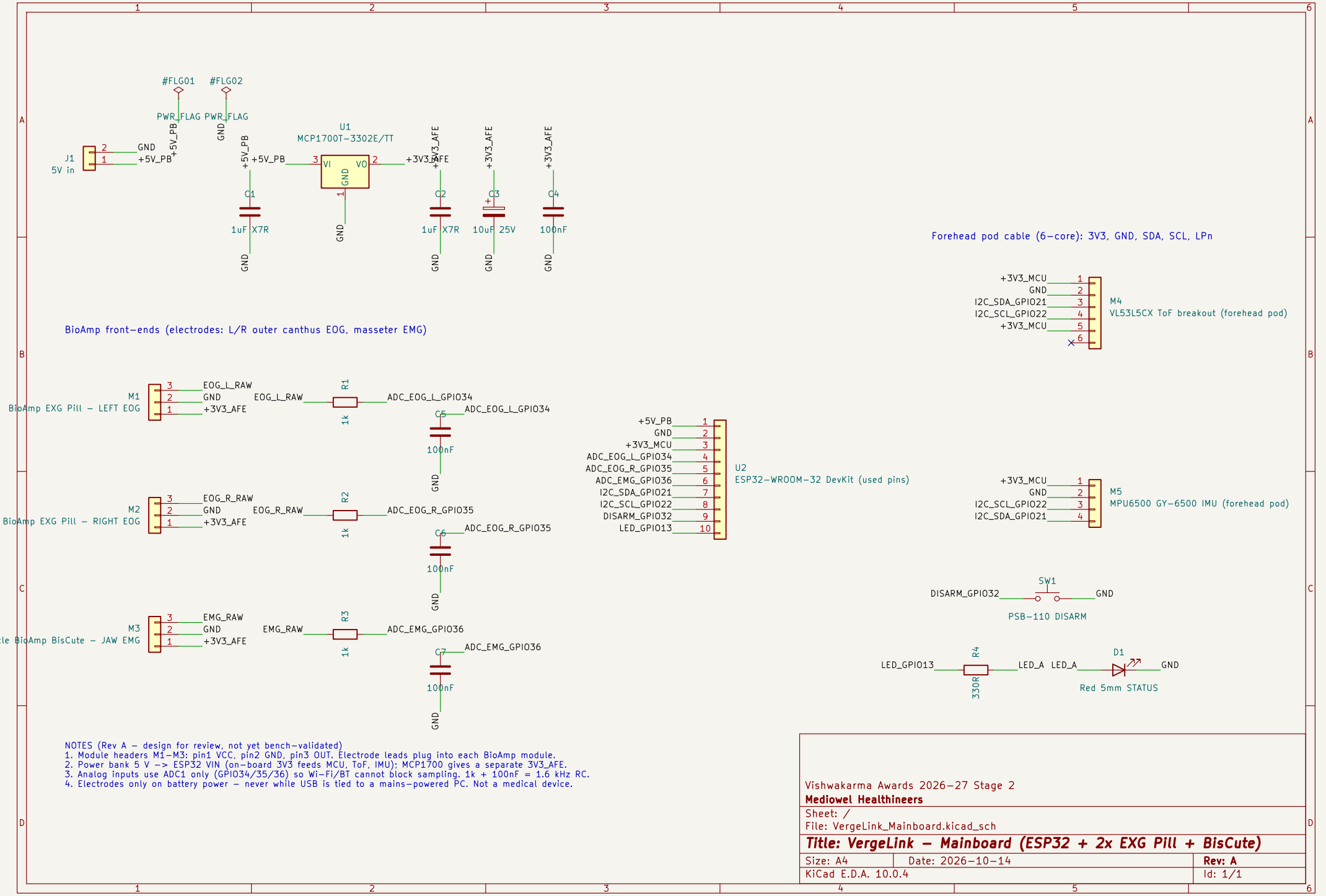

#FLG01
#FLG02
PWR_FLAG
PWR_FLAG
J1
5V in
2
1
GND
+5V_PB
+5V_PB
GND
U1
MCP1700T-3302E/TT
VI
VO
GND
+5V_PB
+3V3_AFE
C1
1uF X7R
C2
1uF X7R
C3
10uF 25V
C4
100nF
GND
BioAmp front-ends (electrodes: L/R outer canthus EOG, masseter EMG)
M1
BioAmp EXG Pill – LEFT EOG
EOG_L_RAW
GND
+3V3_AFE
R1
1k
ADC_EOG_L_GPIO34
C5
100nF
M2
BioAmp EXG Pill – RIGHT EOG
EOG_R_RAW
R2
ADC_EOG_R_GPIO35
C6
100nF
M3
Muscle BioAmp BisCute – JAW EMG
EMG_RAW
R3
ADC_EMG_GPIO36
C7
100nF
U2
ESP32-WROOM-32 DevKit (used pins)
+5V_PB
+3V3_MCU
ADC_EOG_L_GPIO34
ADC_EOG_R_GPIO35
ADC_EMG_GPIO36
I2C_SDA_GPIO21
I2C_SCL_GPIO22
DISARM_GPIO32
LED_GPIO13
Forehead pod cable (6-core): 3V3, GND, SDA, SCL, LPn
M4
VL53L5CX ToF breakout (forehead pod)
I2C_SDA_GPIO21
I2C_SCL_GPIO22
M5
MPU6500 GY-6500 IMU (forehead pod)
SW1
PSB-110 DISARM
R4
330R
D1
LED_A
Red 5mm STATUS
NOTES (Rev A – design for review, not yet bench-validated)
1. Module headers M1–M3: pin1 VCC, pin2 GND, pin3 OUT. Electrode leads plug into each BioAmp module.
2. Power bank 5 V -> ESP32 VIN (on-board 3V3 feeds MCU, ToF, IMU); MCP1700 gives a separate 3V3_AFE.
3. Analog inputs use ADC1 only (GPIO34/35/36) so Wi-Fi/BT cannot block sampling. 1k + 100nF = 1.6 kHz RC.
4. Electrodes only on battery power – never while USB is tied to a mains-powered PC. Not a medical device.
Vishwakarma Awards 2026-27 Stage 2
Mediowel Healthineers
Sheet: /
File: VergeLink_Mainboard.kicad_sch
Title: VergeLink – Mainboard (ESP32 + 2x EXG Pill + BisCute)
Size: A4
Date: 2026-10-14
Rev: A
KiCad E.D.A. 10.0.4
Id: 1/1